Quantification: 543_3
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	13
interior lacuna count	12
frame-edge lacuna count	1

Interior lacunae only: mean, median and sample SD (n = 12)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	2350.2	2188.5	668.925	px^2
major_axis_length_px	lacuna shape	91.5632	84.5141	33.0603	px
minor_axis_length_px	lacuna shape	36.5948	35.0553	10.2044	px
aspect_ratio	lacuna shape	2.6601	2.3867	1.276	unitless
eccentricity	lacuna shape	0.889	0.9075	0.078	unitless
solidity	lacuna shape	0.8742	0.9204	0.1013	unitless
roots_count	headline	2.5	2	2.6799	count
ring_length_r30_px	headline	242.333	258.5	145.763	px
ring_length_r60_px	reported	762.25	806	299.533	px
owned_length_px	ownership-dependent	1993.71	755.541	2420.23	px
edge_count	ownership-dependent	89.75	28.5	109.584	count
mean_edge_length_px	ownership-dependent	17.3649	21.7705	12.0431	px
perimeter_px	normalised	221.015	219.693	100.917	px
ring_area_r30_px2	normalised	8668.5	8576.5	2568.76	px^2
ring_area_r60_px2	normalised	21815.7	21875	4394.01	px^2
in_frame_fraction_r30	normalised	0.9974	1	0.009	unitless
in_frame_fraction_r60	normalised	0.9801	1	0.058	unitless
ring_density_r30	normalised	0.0256095	0.0285228	0.0118545	px^-1
ring_density_r60	normalised	0.0338468	0.0330652	0.00819932	px^-1
roots_per_100px_perimeter	normalised	0.8924	0.8877	0.7902	per 100 px
ring_attached_length_r30_px	network v2	187.583	151	150.982	px
ring_attached_length_r60_px	network v2	476.167	425	361.797	px
ring_length_w_r30_px	network v2	273.552	292.593	167.05	px
ring_length_w_r60_px	network v2	867.258	920.059	338.983	px
sholl_crossings_r10	network v2	5.8333	5	4.3658	count
sholl_crossings_r20	network v2	9.3333	11	4.7162	count
sholl_crossings_r30	network v2	10.4167	10.5	4.814	count
ring_width_mean_r30_px	width	6.8669	7.2188	1.0565	px

Per field
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measure	value	units
analysed_area_px2	1.02406e+06	px^2
total_skeleton_length_px	35600	px
canalicular_length_density_per_px	0.0347635	px^-1
junction_count	869	count
junction_density_per_px2	0.00084858	px^-2
lacuna_count	13	count
lacunae_per_px2	1.269e-05	px^-2
skeleton_component_count	167	count

mean_component_length_px	213.174	px
median_component_length_px	18	px
field_length_density_w_per_px	0.039592	px^-1
field_density_without_flagged_per_px	0.0347635	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	224	px
unattached_component_count	15	count
canalicular_length_density_attached_per_px	0.0347635	px^-1
field_length_density_with_unattached_per_px	0.0349823	px^-1

Canalicular width, the new measure

measure	value	units
width_mean_px	7.7041	px
width_median_px	8	px
width_p10_px	4.4721	px
width_p90_px	10	px
width_pixels_used	35600	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel. Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	2174	80.8669	35.2412	2.2947	0.9	0.9518	0.4109	79.3965	270.898	False
2	3447	90.3677	63.9486	1.4131	0.7066	0.674	0.4223	197.059	916.902	False
3	1883	69.2617	34.8694	1.9863	0.864	0.9448	0.5742	250.833	340.47	False
4	3536	181.867	30.1414	6.0338	0.9862	0.7148	0.0738	327.046	42.4075	False
5	2052	69.2015	39.4228	1.7554	0.8219	0.9112	0.0622	388.671	369.939	False
6	2266	88.1613	35.5691	2.4786	0.915	0.9021	0.1324	473.05	745.191	False
7	1625	80.1248	27.6985	2.8927	0.9383	0.8325	0.2321	545.572	877.076	False
8	1683	71.0123	31.3735	2.2634	0.8971	0.9355	-0.0518	634.68	198.315	False
9	2203	91.2092	31.4053	2.9043	0.9389	0.9295	0.0832	734.301	635.44	False
10	2633	93.5592	36.2778	2.579	0.9218	0.9454	0.0457	780.576	481.495	False
11	531	28.7881	24.8252	1.1596	0.5063	0.9332	0.5942	1012.28	184.834	True
12								458.12	481.03	False
13								857.425	285.616	False

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	2	252	825	184.338	5	36.8676	6.6122
2	9	518	1368	4885.21	223	21.9068	7.3395
3	0	166	570	0	0	0	7.4296
4	5	390	907	6468.61	299	21.6341	7.5552
5	2	265	616	537.073	22	24.4124	7.098
6	5	303	912	2767.86	124	22.3214	7.818
7	1	172	837	2513.11	135	18.6156	6.9851
8	1	108	503	10.8284	1	10.8284	6.7729
9	3	383	1083	5583.47	233	23.9634	7.5455
10	2	265	787	974.009	35	27.8288	7.498
11	0	79	410	0	0	0	6.0343

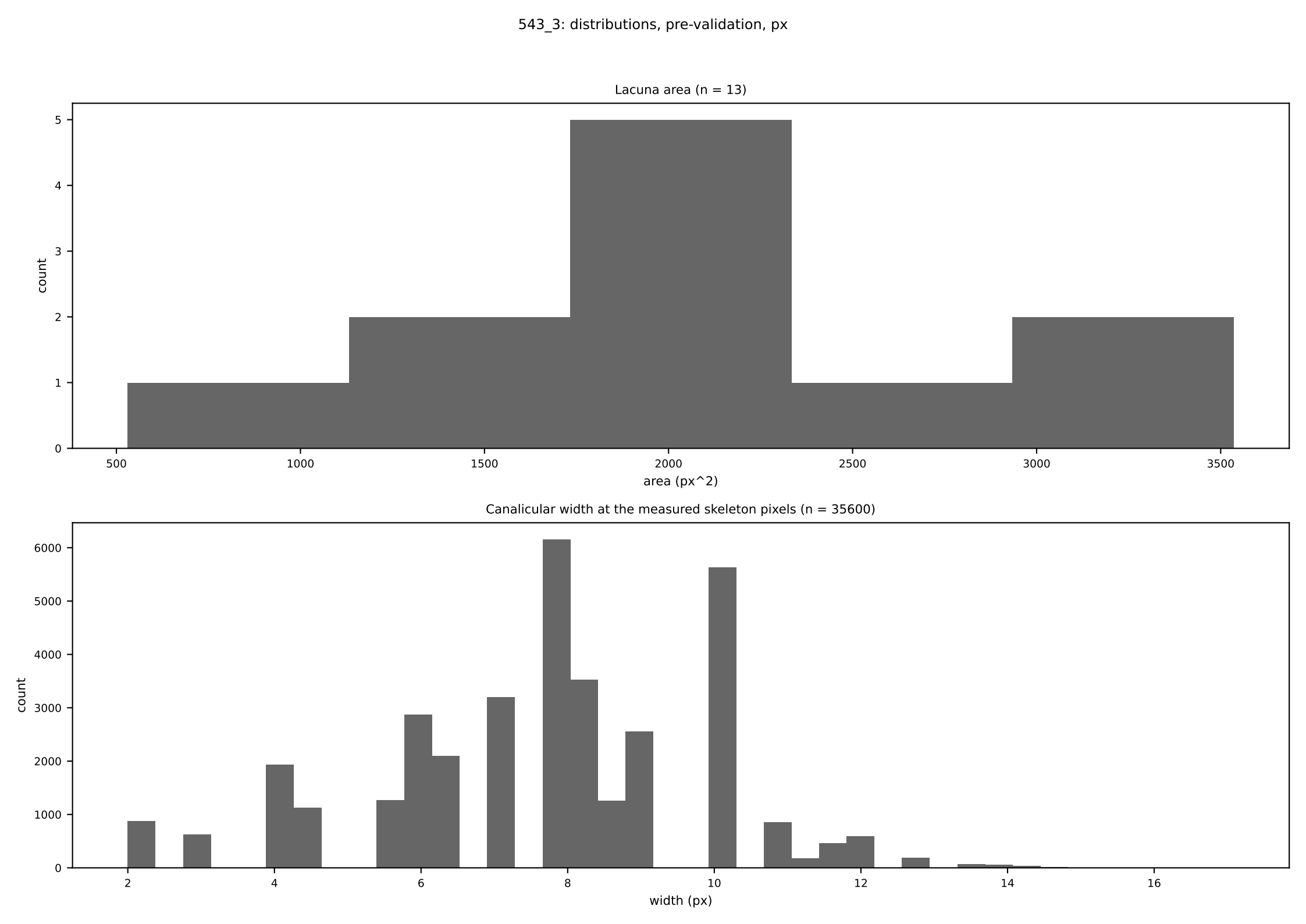
12	0	4	248	0	0	0	4.0322
13	0	82	491	0	0	0	5.7164

Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	210.066	8595	21715	1	0.9616	0.0293194	0.0379922	0.9521
2	363.446	12097	29071	1	1	0.0428205	0.0470572	2.4763
3	184.794	7941	19515	1	1	0.0209042	0.0292083	0
4	415.99	13536	26753	0.9687	0.7992	0.0288121	0.0339027	1.202
5	205.965	8333	20013	1	1	0.0318013	0.03078	0.971
6	241.907	9380	23807	1	1	0.0323028	0.0383081	2.0669
7	229.321	8558	22035	1	1	0.0200981	0.037985	0.4361
8	191.965	7943	21525	1	1	0.0135969	0.0233682	0.5209
9	238.108	9212	24077	1	1	0.0415762	0.0449807	1.2599
10	242.936	9386	24420	1	1	0.0282335	0.0322277	0.8233
11	95.3909	3354	9525	0.6102	0.5725	0.023554	0.0430446	0
12	47.4558	4187	13513	1	1	0.00095534	0.0183527	0
13	80.2254	4854	15344	1	1	0.0168933	0.0319995	0

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	112	318	291.321	958.691	4	12	15
2	469	1167	595.115	1556.15	13	16	19
3	116	384	181.64	645.973	4	7	9
4	379	844	436.191	1029.65	13	13	9
5	238	543	293.865	707.784	7	11	11
6	269	661	338.936	1036.68	7	11	10
7	67	216	192.681	948.181	2	7	11
8	64	153	117.083	561.617	3	4	6
9	333	903	435.948	1220.55	10	14	15
10	186	466	302.764	891.938	6	12	13
11	6	6	84.3553	467.061	1	3	7
12	0	0	3	292.806	0	0	1
13	18	59	94.0833	557.073	1	5	6


543_3: distributions, pre-validation, px
Lacuna area (n = 13)
count
area (px^2)
Canalicular width at the measured skeleton pixels (n = 35600)
count
width (px)